Robot Run Scoring Worksheet

Full 2:30 test runs. Run it like the real match. Write down every run.

A full test run is a practice match: all 8 missions in order, with the **2 minute 30 second** timer running. Each run gets its own run sheet. After a few runs, the summary pages show our **average, best, worst and middle score**, and which missions fail most often.

Our mission order

#	Mission	How it scores (rule book)	Max
1	**Mission 2: Exploding Seeds**	10 for each seed no longer touching the stalk (3 seeds). Keep one for Mission 14.	**30**
2	**Mission 1: Drone Survey**	Drone no longer touching the mat 20 · *bonus:* LiDAR map completely flipped, scan marker partly in the survey area +10	**30**
3	**Mission 3: Flip the Rock**	Research flag down 20 · *bonus:* rock returned to its original starting position +10	**30**
4	**Mission 8: Tangled**	Vine touching the mat 30	**30**
5	**Mission 15: Biocentric Architecture**	Farm dock. Nesting canopy raised 10 · garden skylight completely in 10. (The farm dock bonus is the compost hatch, so no bonus for these two.)	**20**
6	**Mission 6: Leafcutter Frenzy**	Ant touching the nest: 10 for each leaf fragment inside the nest (4 fragments)	**40**
7	**Mission 11: Window to the Past**	Root cover down, touching the mat 20. (Its seed scores only in Mission 14.)	**20**
8	**Mission 14: Seeds of Renewal** if time allows	Each seed contained in the replantation station 5 · *bonus:* also touching the mat +5 each. Two seeds: one from Exploding Seeds, one from Window to the Past.	**20**

Done = the max for that mission. All 8 done = **220**, plus Mission 10 (20), inspection (20) and tokens (50) = **310** total.

Mission 10 habitats: 20
Spider 10 + snail 10 if still in their starting spots. Don't bump them!

Equipment inspection: 20
Robot and all attachments fit in one launch area, under 12 in. tall.

Precision tokens: start with 6
Lose 1 per touch outside home. Left: 6 or 5 = 50, 4 = 35, 3 = 25, 2 = 15, 1 = 10

Jobs for each run

1. **Driver** runs the robot and swaps attachments. **Timer** runs the 2:30 clock and calls out the time. **Scorer** fills in the run sheet. Swap jobs every run.
2. **Reset the field** exactly like the start of a match, with every model in its starting spot.
3. **Go.** When each mission ends, the timer calls out the time left and the scorer writes it in the **Timer** column.
4. **Stop at 0:00.** Missions not tried count as **Failed**, 0 points. If Mission 14 was skipped for time, write "no time" in its notes.
5. **Score each mission,** add up the total, and agree on the **one change** to make before the next run.

✓ **Done** full points for that mission

½ **Partial** some points, not all

✗ **Failed** 0 points

Rules that keep the numbers honest

- **Never skip a bad run.** If we only write down good runs, our average lies to us.
- **Change one thing between runs** (code, attachment or how we line up), so we know what made the difference.

Run Sheet Run # _____

Date: _____ **Driver:** _____ **Timer:** _____ **Scorer:** _____

#	Mission	Result	Points	Timer at end	What happened? What should we change?
1	MISSION 2 **Exploding Seeds** 10 per seed off the stalk (3 seeds)	☐ ✓ Done ☐ ½ Partial ☐ ✗ Failed	of 30	m : ss left	
2	MISSION 1 **Drone Survey** Drone off mat 20 · bonus: LiDAR map flipped +10	☐ ✓ Done ☐ ½ Partial ☐ ✗ Failed	of 30	m : ss left	
3	MISSION 3 **Flip the Rock** Flag down 20 · bonus: rock back in its spot +10	☐ ✓ Done ☐ ½ Partial ☐ ✗ Failed	of 30	m : ss left	
4	MISSION 8 **Tangled** Vine touching the mat 30	☐ ✓ Done ☐ ½ Partial ☐ ✗ Failed	of 30	m : ss left	
5	MISSION 15 **Biocentric Architecture** Farm dock: canopy up 10 · skylight in 10	☐ ✓ Done ☐ ½ Partial ☐ ✗ Failed	of 20	m : ss left	
6	MISSION 6 **Leafcutter Frenzy** Ant at nest: 10 per leaf fragment (4)	☐ ✓ Done ☐ ½ Partial ☐ ✗ Failed	of 40	m : ss left	
7	MISSION 11 **Window to the Past** Root cover down 20 · grab its seed for M14	☐ ✓ Done ☐ ½ Partial ☐ ✗ Failed	of 20	m : ss left	
8	MISSION 14 · IF TIME ALLOWS **Seeds of Renewal** In station 5 each · also touching mat +5 each (2 seeds)	☐ ✓ Done ☐ ½ Partial ☐ ✗ Failed	of 20	m : ss left	

MISSION POINTS	+ MISSION 10 HABITATS	+ INSPECTION	TOKENS LEFT	+ TOKEN POINTS	= TOTAL POINTS
of 220	0, 10, 20	0 or 20	of 6	max 50	of 310

TOTAL TIME USED	✓ DONE	½ PARTIAL	✗ FAILED
m : ss	of 8	of 8	of 8

Mission 10: 10 for each habitat (spider, snail) still in its starting spot at the end. **Token points:** 6 or 5 left = 50, 4 = 35, 3 = 25, 2 = 15, 1 = 10. **Total time used** = 2:30 minus the timer when the robot finished its last mission (2:30 if the timer hit 0:00).

The ONE change for the next run

Who will do it: _____ **Did it help? (next run)** _____

All Runs: Points and Time

Copy the totals from each run sheet into one row. The first row is an example.

Run	Date	Total points	Time used	✓ Done	½ Partial	✗ Failed	Tokens left	The change we made after this run
eg	*11/12*	*145*	*2:22*	*3*	*2*	*1*	*5*	*Slow down the turn before Mission 6*
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								

Our points numbers

Use only the **Total points** column, and don't count the example row.

Average

Add up all the totals, then divide by the number of runs.

_____ pts

Max (best)

The biggest total.

_____ pts

Min (worst)

The smallest total.

_____ pts

Median (middle)

Put the totals in order. The middle one. Even number of runs? Average the two middle ones.

_____ pts

Work space: totals in order, smallest to biggest

What the numbers tell us: is the average close to the max? If not, which missions keep dragging it down?

Why the median? One really bad run (robot fell over) pulls the average down a lot but barely moves the median. If the average is much lower than the median, we have a few disaster runs to look at.

Mission by Mission

For each run, write the mission's result: ✓, ½ or ✗, and its points underneath if there is room. Then count across. The mission with the most ✗ is the one to fix first.

Mission	1	2	3	4	5	6	7	8	9	10	11	12	✓ count	✗ count
1 · Mission 2 Exploding Seeds														
2 · Mission 1 Drone Survey														
3 · Mission 3 Flip the Rock														
4 · Mission 8 Tangled														
5 · Mission 15 Biocentric Architecture														
6 · Mission 6 Leafcutter Frenzy														
7 · Mission 11 Window to the Past														
8 · Mission 14 Seeds of Renewal														
Not bumped · Mission 10 Fragile Microhabitats														

What to fix next

#	Mission	Why does it fail or only get partial points?	Fix and who
1			
2			
3			

Time check

Slowest mission (biggest time gap on the run sheets): _____

Average time used: _____ of 2:30

How often did we have time for Mission 14? _____ out of _____ runs

The time a mission takes is the gap between the timer reading before it and after it. The first mission (Exploding Seeds) starts at 2:30.